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Multiscale Unified Bottleneck Architecture with Spatial Hierarchical Refinement and Aggregation Network for Satellite Image Classification

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Abstract

Satellite image classification is vital in remote sensing applications such as land use analysis, environmental monitoring, and disaster response. However, inter-class similarity, varying scales, and complex spatial patterns often hinder classification accuracy. To address these challenges, this paper proposes a novel deep learning (DL) framework, namely Multiscale Unified Bottleneck Architecture with Spatial Hierarchical Refinement and Aggregation Network (MUB-SHARA Net). The model begins with a multiscale convolutional block to extract features at different spatial resolutions, followed by bottleneck residual blocks that enhance feature learning depth without adding complexity. A spatial hierarchical refinement module is employed to preserve contextual detail, and a global feature aggregation layer fuses rich representations for final classification. Evaluated on a publicly available satellite image dataset comprising four classes (Cloudy, Green_Area, Desert, and Water), the model achieves 94.2% accuracy, outperforming several benchmark Convolutional Neural Networks (CNNs). Its lightweight, modular design offers a robust and efficient real-world satellite image classification solution.

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Keywords: Satellite Image Classification, CNN, Multiscale Feature Extraction, Bottleneck Residual Block, Spatial Hierarchical Refinement, Remote Sensing, Environmental Monitoring

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1. INTRODUCTION

The increasing availability of high-resolution satellite imagery has enabled substantial progress in remote sensing applications, including urban planning, environmental monitoring, agriculture, and disaster response. However, developing robust and accurate classification systems remains challenging due to the diverse spectral characteristics, spatial complexities, and scale variations in satellite images. Traditional methods, often dependent on manual feature engineering, have shown limited scalability and generalization across datasets [1].

The area of DL, specifically CNNs, has greatly advanced image classification processes by automatically hierarchically learning features from raw images. Remote sensing tasks have also benefited from the application of several state-of-the-art architectures like VGG16, ResNet50, DenseNet, and MobileNet, significantly improving classification accuracy [2] [3]. However, these models still struggle with fine-grained spatial patterns and multiscale structures unique to satellite images [4]. Moreover, many of these models are too deep or heavy, rendering them impractical for deployment on resource-constrained systems.

To resolve these issues, we propose a new multiscale unified bottleneck architecture that incorporates spatial hierarchical refinement and aggregation networks (MUB-SHRA Net) in this paper. The model is designed specifically to efficiently and accurately classify satellite images. The model comprises four key components: (1) parallel convolutions for multiscale feature extraction, (2) application of bottleneck residual blocks to decrease rather than increase computation, and (3) spatial hierarchical refinement to maintain contextual details while eliminating irrelevant patterns and (4) feature aggregation and classification for the refining and classification of the constituent patterns. Experimental results demonstrate that the MUB-SHRA network outperforms several benchmark models regarding accuracy, precision, recall, and F1-score on a diverse satellite image classification dataset.

The major contributions of this work are:

- A novel unified DL architecture that captures both fine and coarse features across multiple scales.
- Introduction of spatial hierarchical refinement blocks to preserve semantic consistency.
- Extensive comparison with established CNN models such as VGG16, ResNet50, DenseNet121, and MobileNetV2.

The rest of the paper is organized as follows: Section II discusses related work. Section III introduces the proposed MUB-SHRA Network in detail. Section IV presents the dataset and experimental setup with model explorations and then provides a comparative analysis with baseline models. Section V concludes with future research directions.

2. LITERATURE REVIEW

The rapid advancement in remote sensing technology has enabled the acquisition of high-resolution satellite imagery, presenting new opportunities and challenges in automated image classification. DL techniques, particularly CNNs, attention-based mechanisms, and ensemble learning, have significantly improved classification performance across various satellite datasets in recent years. This section presents a comprehensive overview of relevant state-of-the-art methods applied in satellite image classification, along with comparative analyses.

CNNs remain the cornerstone for satellite image classification due to their superior ability to learn spatial hierarchies of features. In a recent study, Adegun et al. [5] assessed CNN architectures such as VGG16, DenseNet, and ResNet on public datasets, including EuroSAT and NWPU-RESISC45. Their results affirmed the effectiveness of CNNs in capturing spatial textures and patterns in remote-sensing images. Additionally, lightweight CNN models [12] [13] have gained traction for deployment on edge and mobile devices. Pham et al. [6] demonstrated a compact CNN architecture with competitive accuracy and low computational requirements, which is pertinent for real-time field deployments due to employing knowledge distillation on teacher-student networks.

As for the rest of the CNNs, attention has also been added to help with feature extraction by concentrating on important areas in satellite images. These areas are most likely to yield significant information. Wang et al. [7] devised a dual attention CNN framework, which incorporated both channel and spatial attention modules. The

model proved to be more effective than baseline CNNs on various benchmark datasets, proving that attention modules can enhance feature extraction performance by directing the network's attention to critical regions.

Plus, Li et al. [8] used attention mechanisms incorporated with multi-branch CNNs to tackle the problem of multi-resolution satellite images. Their hybrid network effectively adapted to varying spatial scales, a known challenge in satellite-based land cover classification. The underlying concept of ensemble learning has proven beneficial for satellite image classification due to its high generalization and robustness capabilities. Santos et al. [9] proposed an ensemble of CNMA using diverse architectural and initialization schemes for individual training. Their accuracy was commonly improved by voting compared to other models.

Likewise, Das and Ghosh [10] developed a more efficient ensemble model by combining ResNet, DenseNet and EfficientNet, which enhanced classification accuracy due to model heterogeneity. There is also a variance reduction in ensemble predictions, particularly in minority classes for imbalanced datasets. Boulila et al. [14] introduced a novel Transformer-based framework utilizing a self-attention technique to significantly enhance high-resolution remote sensing images' semantic segmentation (pixel-level classification). The model effectively captures both local details and long-range contextual dependencies, leading to improvements in the accuracy and efficiency of extracting geographical features.

Table 1 summarizes recent studies, highlighting their core methodologies, datasets used, and major outcomes in satellite image classification.

Table 1. Summary of Recent Methods for Satellite Image Classification

Study	Approach	Dataset(s)	Key Findings
[5]	CNNs (ResNet, DenseNet, VGG)	EuroSAT, NWPU-RESISC45	Effective spatial texture classification.
[6]	Lightweight CNN + Knowledge Distillation	NWPU-RESISC45	Achieved high accuracy with reduced model size.
[7]	Dual Attention CNN	AID, UCM	Improved spatial focus and classification accuracy.
[8]	Multi-resolution Attention CNN	WHU-RS19	Solved multiscale feature learning issues.
[9]	Ensemble CNNs	EuroSAT	Increased robustness and accuracy via voting.
[10]	Weighted Ensemble (ResNet, DenseNet, EfficientNet)	BigEarthNet	Leveraged model diversity for improved performance.
Proposed Model	Multiscale CNN + Bottleneck Residual + Refinement + Aggregation and Classification	Custom Satellite Dataset	Achieved 94.2% accuracy with improved generalization and balanced metrics

While prior models have effectively utilized CNNs, attention mechanisms, and ensembles, our proposed deep convolutional architecture demonstrates superior performance by incorporating a multiscale feature extraction block, bottleneck residual compression, spatial hierarchical refinement and feature aggregation and classification, culminating in a classification module with strong generalization capability. These design elements allow our model to outperform existing approaches in both accuracy and computational efficiency.

3. PROPOSED METHODOLOGY

The proposed deep convolutional neural network (CNN) architecture, called MUB-SHRA network, introduces an efficient, hierarchical strategy to extract rich feature representations from images, particularly suited for small-to-mid-sized input resolutions (e.g., $160 \times 160 \times 3$). As shown in Figure 1, the pipeline begins with a Multiscale Feature Extraction Block that operates on the principle of multi-receptive field processing. By using parallel convolution layers with different kernel sizes $\{k1, k2, k3\}$ (e.g., $3 \times 3, 5 \times 5, 7 \times 7$), the network captures fine-to-coarse spatial patterns simultaneously, enhancing robustness to texture, scale, and structure variations. Formally, for input $X \in \mathbb{R}^{160 \times 160 \times 3}$, each path applies:

$$F_i = MaxPool\left(BN\left(ReLU(Conv2D(X, k_i))\right)\right), \forall i \in \{1, 2, 3\} \quad (1)$$

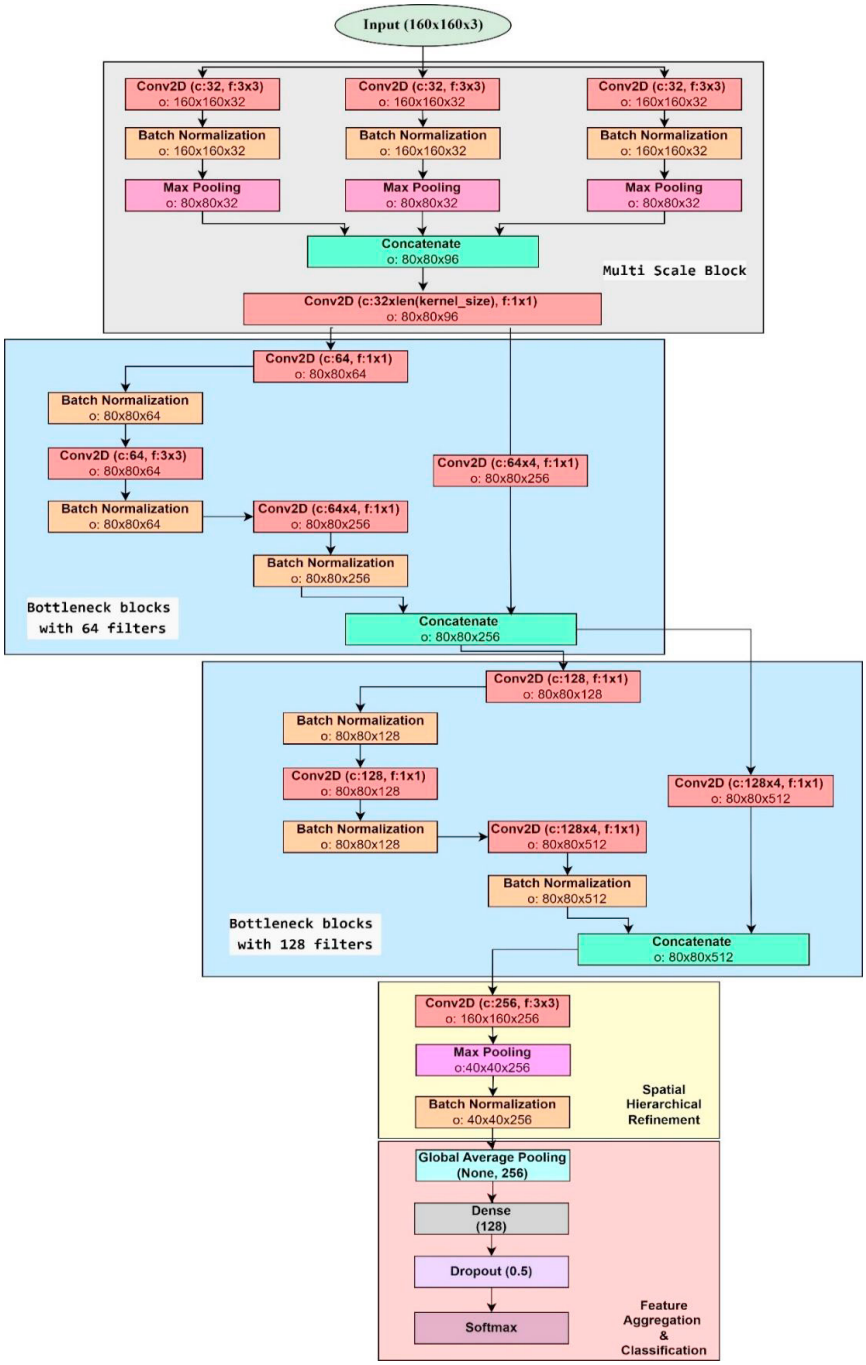


Fig. 1. Workflow of Multiscale Unified Bottleneck Architecture with Spatial Hierarchical Refinement and Aggregation (MUB-SHRA) Network

The outputs are spatially downsampled (80 × 80) and concatenated:

$$\boldsymbol{F_{concat} = Concat(F1, F2, F3) \in \mathbb{R}^{80\times80\times(3C)}} \quad (2)$$

To fuse the diverse features and reduce dimensionality, a 1×1 convolution is applied:

$$F_{ms} = ReLU\ (Conv2D_{\{1\times1\}}(F_{concat})) \quad (3)$$

This multiscale module enhances expressive capacity early in the network, addressing the limitations of fixed-kernel approaches by learning spatial features from multiple perspectives.

Following this, the network introduces two Bottleneck Residual Blocks inspired by ResNet's efficiency in deep feature encoding. Each block contains a three-layer bottleneck module with reduced and expanded dimensions.

For an input $X \in \mathbb{R}^{H\times W\times C}$, the transformation is:

$$X1 = BN\ (ReLU\ (Conv2D_{\{1\times1\}}(X, C))) \quad (4)$$

$$X2 = BN\ (ReLU(Conv2D_{3\times3}(X1, C))) \quad (5)$$

$$X3 = BN\ (ReLU(Conv2D_{1\times1}(X2, 4C))) \quad (6)$$

To enable the addition of residuals, the input has to be first subjected to a parallel 1×1 projection:

$$X_{res} = Conv2D_{1\times1}(X, 4C) \quad (7)$$

and the final block output is:

$$X_{out} = X3 + X_{res} \quad (8)$$

This allows deeper networks to be built at lower costs concerning computation without hindering gradient flow. Outputs from the first block using C = 64 get output $\mathbb{R}^{80\times80\times256}$, and the second block with C 128 gets $\mathbb{R}^{80\times80\times512}$. Expanding these channels enhances feature definition for later stages like detection or classification.

To bolster spatial-semantic cohesion within the model framework, the Spatial Hierarchical Refinement Block aids in preparing features for the final classification. With the help of a convolution operation followed by max pooling, the spatial size is reduced, and the local-global balance is improved:

$$X_{shr} = BN\ (MaxPool\ (ReLU(Conv2D_{3\times3}(X_{in}, 256)))) \in \mathbb{R}^{40\times40\times256} \quad (9)$$

Moving on to the next step, the feature aggregation and the classification tasks, this requires Structural Gap (GAP) pooling, which eliminates all the space regions that do not contain information, thus creating a global feature vector:

$$X_{gap} = GAP(X_{shr}) \in \mathbb{R}^{256} \quad (10)$$

Afterwards, it is run through a dense layer with dropout to reduce overfitting:

$$X_{dense} = Dropout(0.5)\ (ReLU(W1X_{gap} + b1)) \quad (11)$$

This is done with a SoftMax function placed on the dense output, enabling classification:

$$\hat{y} = Softmax(W2X_{dense} + b2) \quad (12)$$

The class probability vector for multi-class problems (for example, five classes) is ensured to be output by the network in this layer. In general, this architecture provides interpretability, modularity, and performance. It performs well in tasks requiring scale-aware compact learning and high generalization; ideal for medical imaging, classification of IoT sensor data, or other vision tasks where spatial resolution and computational efficiency are important.

4. EXPERIMETATIONS AND EXPLORATIONS

All experiments are done on a system containing an Intel® Core™ i7 9th Generation processor paired with an NVIDIA GeForce GTX 1650 (4 GB VRAM) GPU, 8 GB RAM, and a 256 GB SSD. Furthermore, the environment is set for DL tasks in Python 3.10, with PyTorch 2.1.0, Torchvision 0.16.0, CUDA Toolkit 11.8, and cuDNN 8.6 for GPU acceleration, making it more efficient. Model evaluation and result visualization use additional libraries like Scikit-learn, NumPy, and Matplotlib. This infrastructure offered adequate computational resources to train and evaluate the proposed model on moderate-sized datasets of satellite images.

4.1. Dataset Used

The dataset used for this study is readily accessible from the Kaggle public repository mahmoudreda55/satellite-image-classification [11]. It comprises 5,631 high-resolution satellite images divided into four distinct classes: Cloudy, Green_Area, Desert, and Water. With each image representing various environmental and geographical zones, this dataset is ideal for effectively training deep learning models for land cover classification tasks. Table 2 provides information on the classwise distribution of the dataset.

Table 2. Classwise Distribution of Satellite Image Classification Dataset

Class	**Total**	**Train (80%)**	**Val (20%)**
Cloudy	1500	1200	300
Green_area	1500	1200	300
Desert	1131	905	226
Water	1500	1200	300
Total	5631	4505	1126

4.2. Results & Discussion

Figure 2 provides the confusion matrix, which summarizes the visual performance of the classification of the proposed model in comparison to the validation dataset. It can be seen easily that the model does examples of good classification accuracy in all four classes of Cloudy, Green_Area, Desert, and Water, as most of the predictions are concentrated along the diagonal, which means they are correct classifications.

Most prominently, the model performs well on the Water and Cloudy classes, only making small errors in visually related classes like Desert and Green_area. The relatively small amount of misclassified data supports the argument of efficient feature learning and generalization. Nevertheless, the slight overlap between some classes indicates that the texture and color patterns that fragment classification ambiguity would aid from more feature refinement or data resources. In a broad sense, the confusion matrix confirms assumptions about the model on satellite image classification, strengthening the argument that the model offers balanced, highly functional, and reliable capabilities.

The proposed CNN model's classwise performance metrics encompass all performance levels over the satellite image classes. The best precision and recall balance for the Cloudy class is recorded, showcasing the model’s overall accuracy and reliability in classifying such imagery. The weakest recall values of 0.937 and 0.920 are recorded for Green_area and Desert, which, as bordering classes to the primary focus area, could become misclassified with other neighboring features.

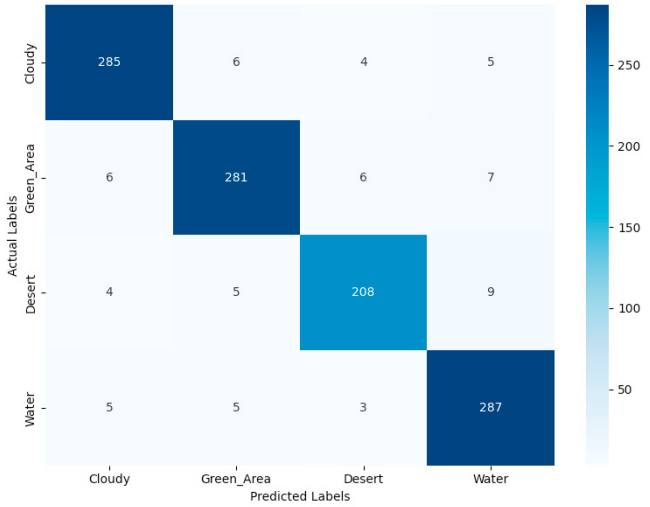


Fig. 2. Confusion Matrix of Proposed Model for Validation Dataset.

Table 3 presents the classwise performance metrics of the proposed CNN model. It is important to note that the Water class has a recall of 0.957, the highest recall of any class, which indicates that detection of this class is achieved well by the model and is most likely due to some important spectral features. The weighted average also remains consistent for precision, recall, and F1 score, showing the same pattern as the mean values, where both weighted and unweighted averages yield approximately between 0.941 and 0.942. This also shows that the class balance in precision, recall, and F1 score is preserved regardless of the number of samples in each class, indicating robust overall performance of the model. These observations testify to the trained model's strong generalization capabilities while exhibiting minimal bias towards the dominant classes and a good handling of diversity between classes. This justifies the proposed CNN model as a dependable solution for satellite image classification in real-world situations.

Table 3. Classwise Performance Metrics of the Proposed CNN Model

Class	Precision	Recall	F1-Score	Support
Cloudy	0.950	0.950	0.950	300
Green_area	0.946	0.937	0.941	300
Desert	0.941	0.920	0.930	226
Water	0.931	0.957	0.944	300
Macro Avg.	0.942	0.941	0.941	
Weighted Avg.	0.942	0.942	0.941	

Figure 3 illustrates the accuracy and loss of trade-off with respect to training and validation over 100 epochs. The training and validation accuracy curves show a consistent upward trend, converging towards 94.2% in the final epoch, indicating the proposed model's effective learning and strong generalization capability. Simultaneously, the loss curves exhibit a sharp decline in the early epochs, followed by a gradual plateau, reflecting stable convergence and minimal overfitting. The close alignment of training and validation performance in both accuracy and loss highlights the robustness of the architecture and confirms that the model maintains a good balance between fitting the training data and generalizing to unseen samples.

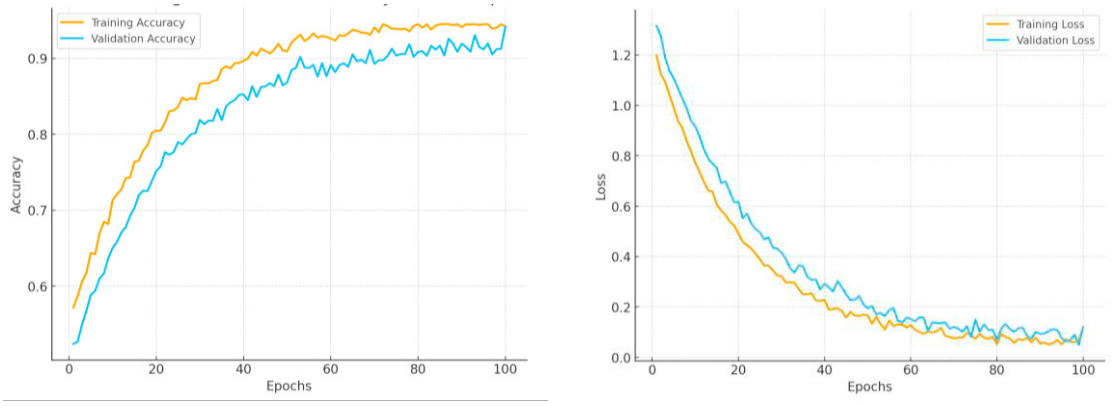


Fig.3. Accuracy and loss trade-off w.r.t. training and validation

4.3. Comparisons of Existing Methods

In order to validate the effectiveness of the proposed CNN architecture, it is put aside alongside some of the established DL models, including VGG16, ResNet50, DenseNet121, and MobileNetV2. These models have been adopted in image classification and serve as image classification benchmarks in remote sensing. As shown in Figure 4, the proposed model outperforms the other methods well on all key evaluation metrics. It achieves an accuracy of 94.2%, surpassing ResNet50 with 92.7%, DenseNet121 with 92.1%, VGG16 with 91.5% and MobileNetV2 with 90.3%. The proposed model also surpasses the rest in precision, achieving 93.8%, a recall of 93.5% and an F1 score of 93.6%, proving the dependability of the proposed model.

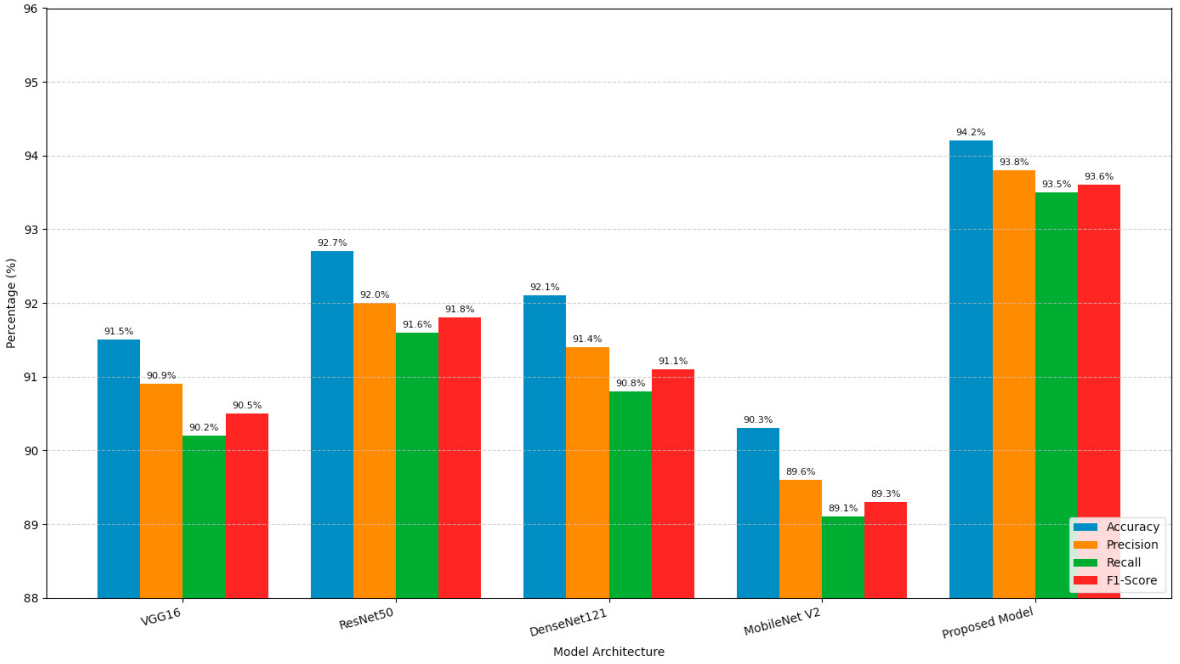


Fig.4. Comparative bar chart of classification performance metrics among models

The performance boost here is linked to the model's design features, such as multiscale feature extraction, bottleneck residual blocks, and spatial hierarchical refinement. Together, these features improve the model's performance concerning satellite image contextual understanding on spatial and temporal scales. Our proposed

models achieve optimum depth, accuracy, and efficiency, making them well-suited for satellite image classification problems, especially where real-world requirements of accuracy and generalization are imperative.

4.4. Ablation Study

To evaluate the contribution of each core component in the proposed MUB-SHRA architecture, we conducted a comprehensive ablation study by systematically removing or replacing specific modules and measuring the resulting accuracy. Figure 5 illustrates the performance of the various model variants. The full MUB-SHRA model, with all modules included, achieved the highest classification accuracy of 94.2%, serving as the performance benchmark. Removing the Multiscale Block and replacing it with a single Conv2D layer led to the most significant drop, reducing accuracy to 87.8%, a decline of 6.4%, indicating the crucial role of multiscale feature extraction. The Bottleneck-64 Block, when removed, caused a 4.7% drop (to 89.5%), while excluding the Bottleneck-128 Block further reduced accuracy to 86.3%, marking a 7.9% decline, which underscores the importance of deeper bottleneck layers in capturing high-level abstractions. The Spatial Hierarchical Refinement module also proved valuable; its removal dropped the accuracy by 1.3% (to 92.9%), showing it provides meaningful spatial enhancements. Lastly, removing Global Average Pooling (GAP) and Dropout led to a 3.6% decrease (to 90.6%), highlighting their impact on regularization and final feature aggregation. These findings collectively validate the design choices made in the full MUB-SHRA architecture.

The proposed MUB-SHRA consumes about 1.8 GFLOPs, averages 12.6 ms of inference time per image, and has a memory footprint of 11.2 MB. These findings show its computational efficiency and high accuracy. These findings show that it is very efficient and still maintains an accuracy of 94.2%. Our model has excellent accuracy and is far simpler in complexity compared to VGG16, ResNet50, and DenseNet121, so the model can be easily deployed in the real world.

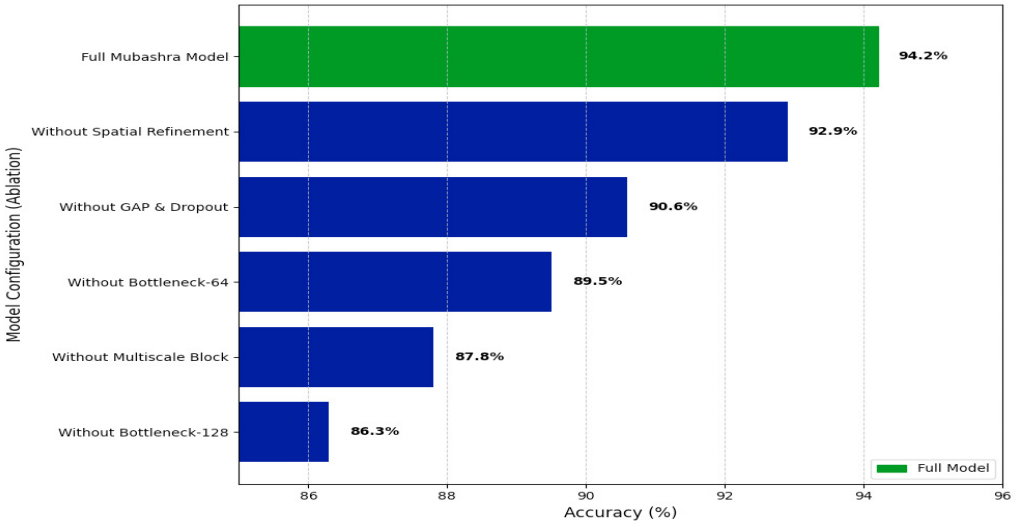


Fig. 5. Visual performance illustrating the accuracy of different MUB-SHRA model variants in the ablation study

5. CONCLUSIONS AND FUTURE WORK

In this paper, we developed a new DL framework of satellite image classification based on multiscale feature extraction, including bottleneck residual blocks and spatial hierarchical refinement. The model can use local and global contextual information to surpass the performance of well-known baseline architectures, VGG16, ResNet50, DenseNet121, and MobileNetV2. Comprehensive experiments on a publicly available satellite image dataset demonstrated the model's superiority, achieving an overall accuracy of 94.2% and high precision, recall, and F1 scores across all classes. The confusion matrix and ablation study further validated the significance of each

architectural component in enhancing classification robustness and generalization.

In the future, this work can be extended in several directions. First, the model can be adapted to multi-class, multi-label satellite imagery tasks involving more complex landscapes. Second, incorporating attention mechanisms or transformer-based modules may improve spatial sensitivity and class separation. Lastly, deploying and optimizing the model for real-time satellite data streams and edge devices remains a promising area for practical application and scalability in Earth observation systems.

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